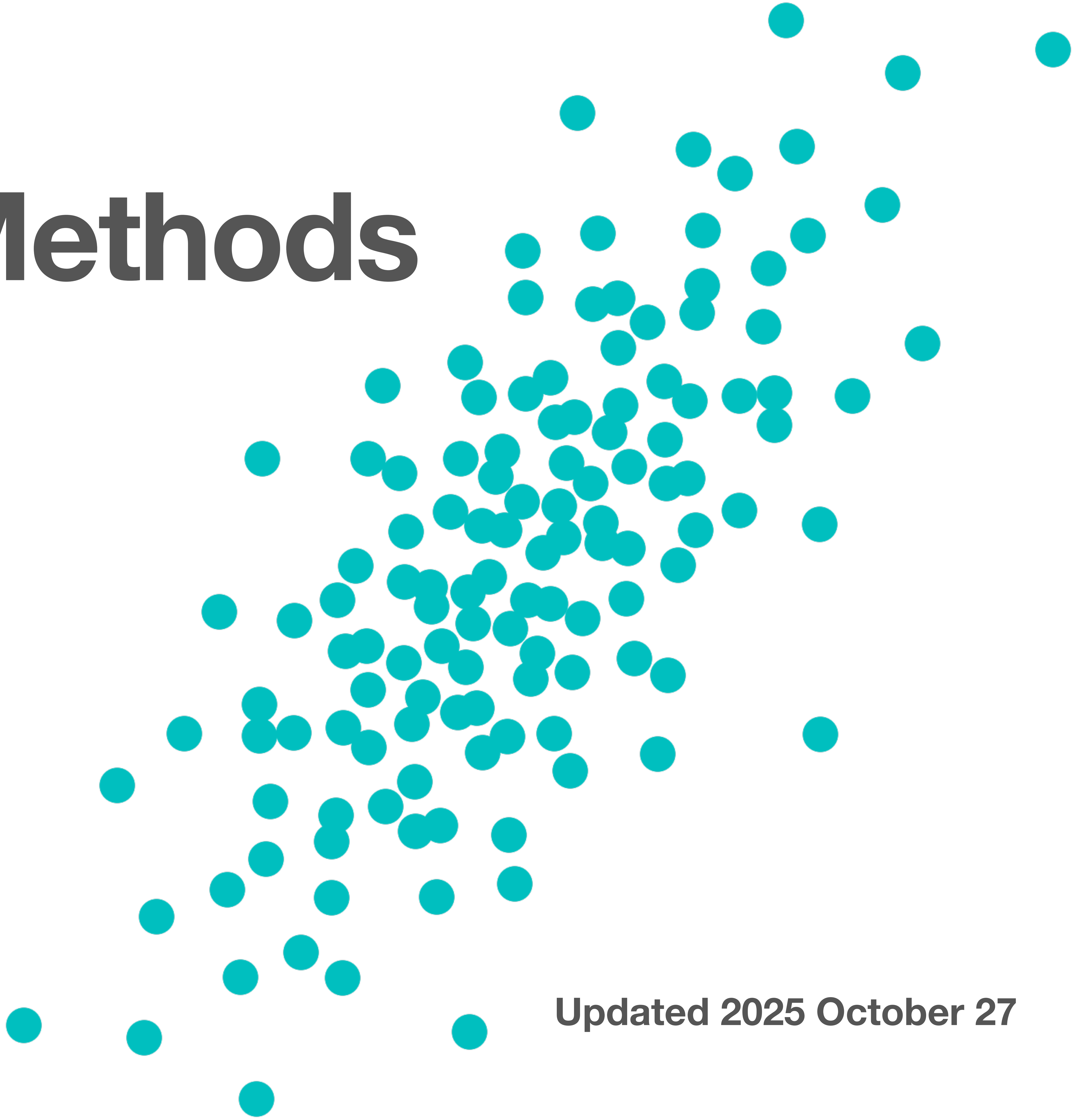
Monte Carlo Methods

Introduction
Generating Samples
Markov Chain Monte Carlo
Improving Efficiency
Selected Topics

Git website and repository
Canvas



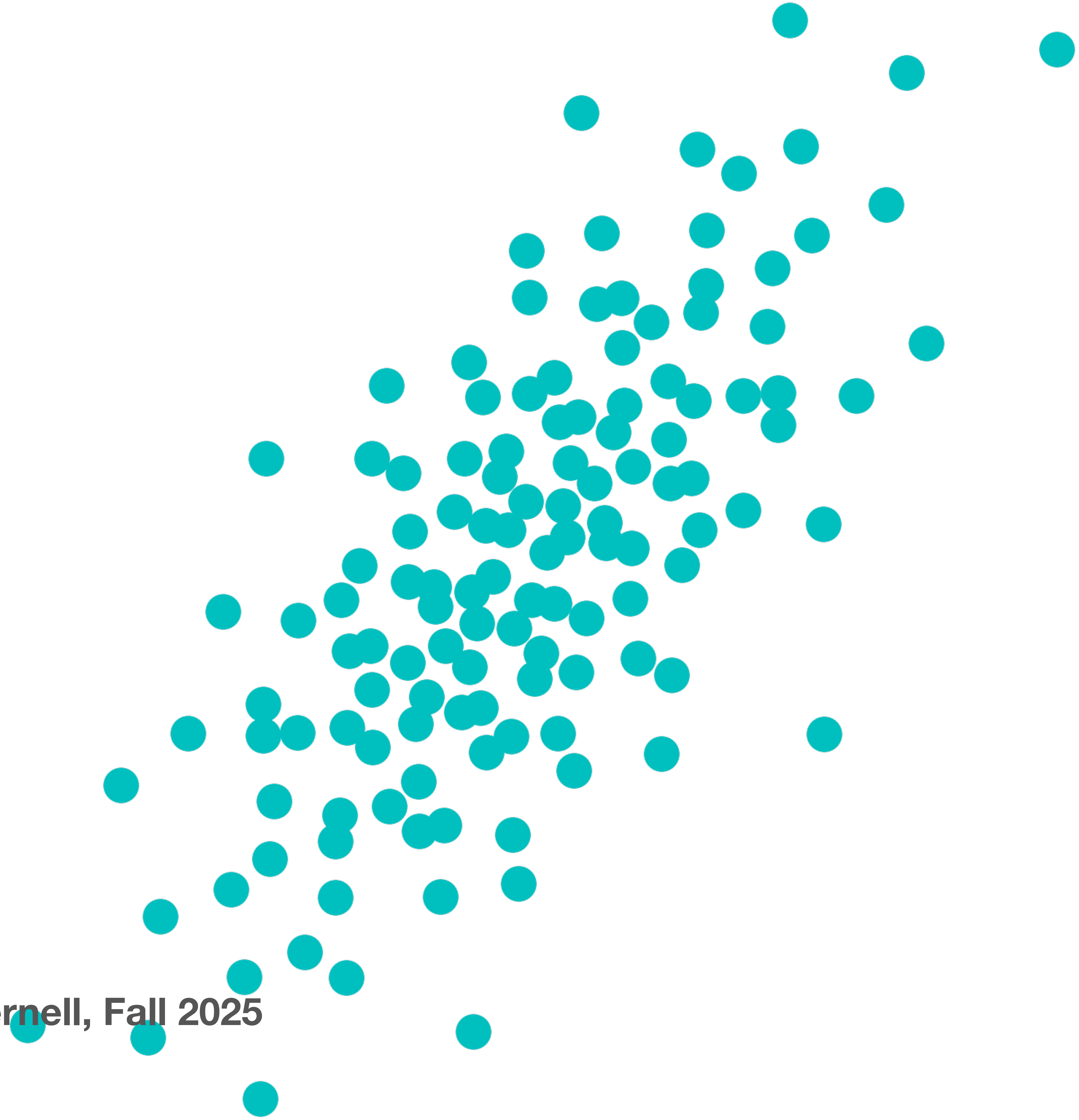
Fred Hickernell, Fall 2025

Updated 2025 October 27

Introduction

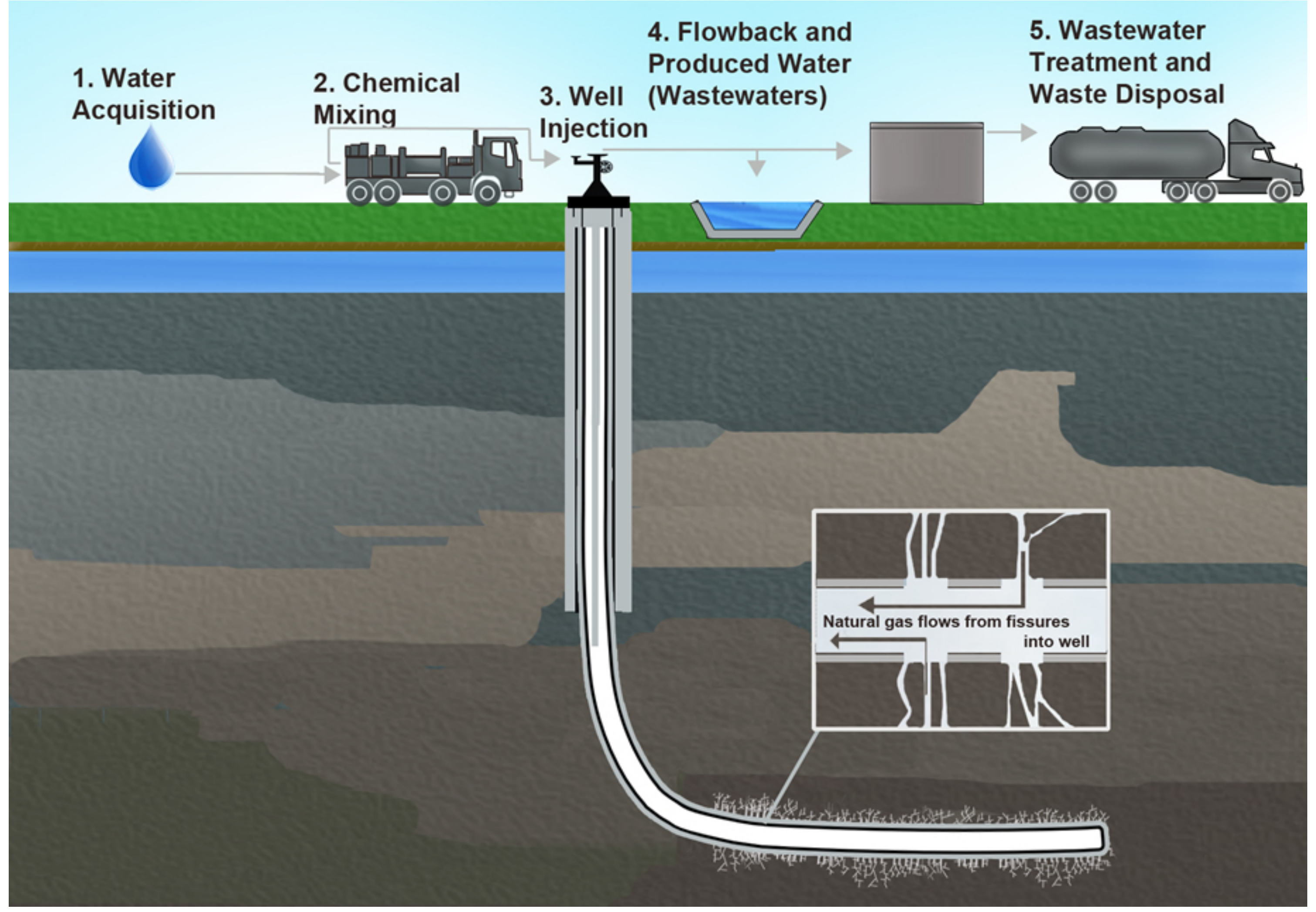
Owen, Chapters 1 & 2

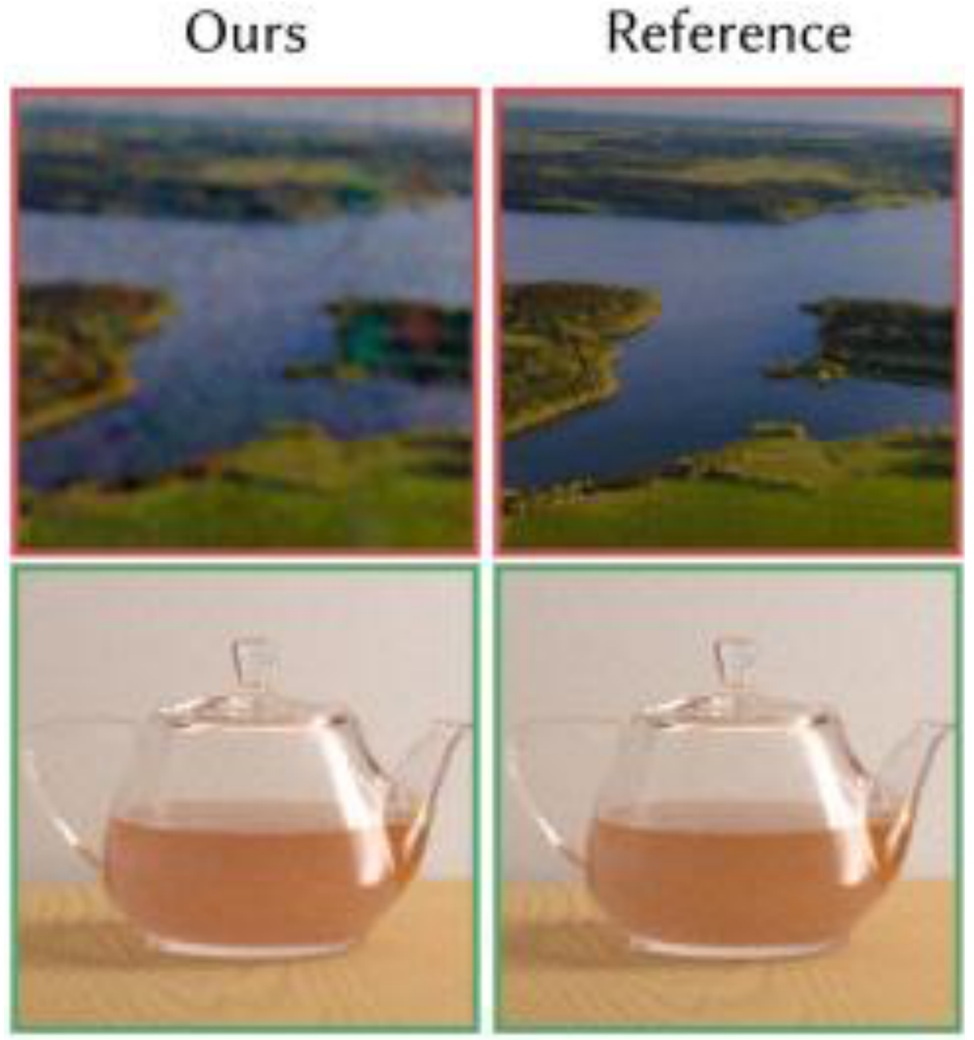
Assignment 1 Due Sep 5

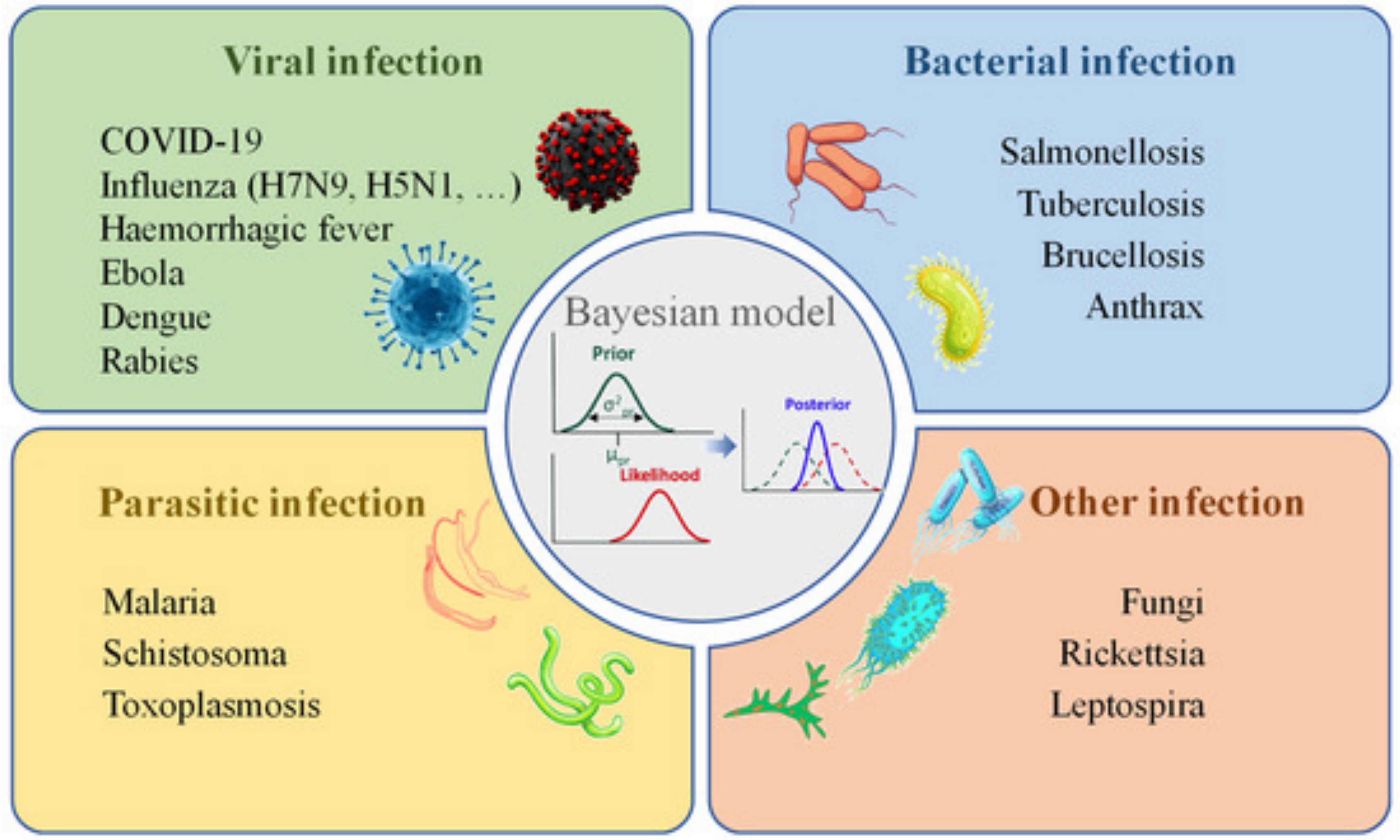


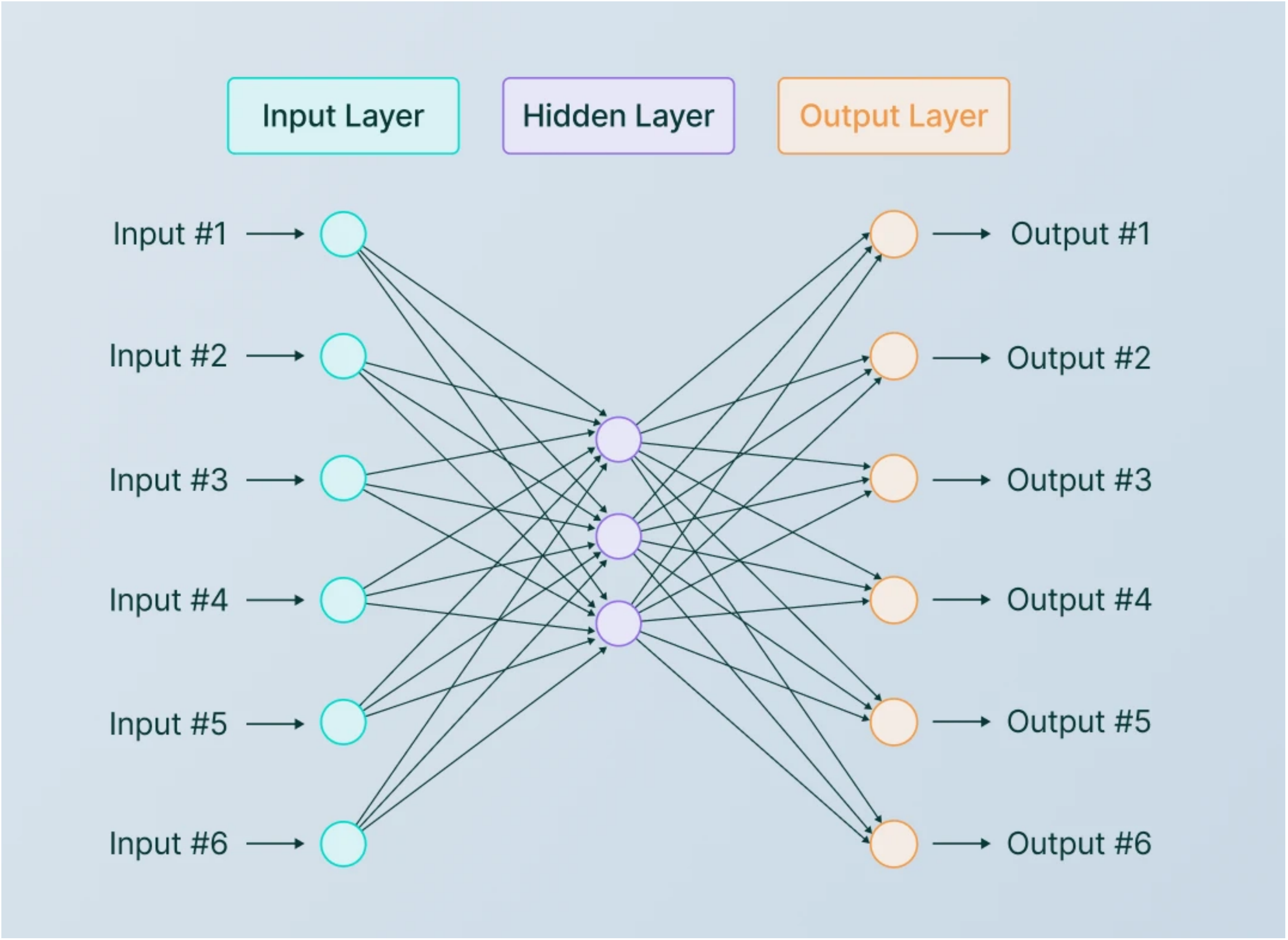
MATH 565 Monte Carlo Methods, Fred Hickernell, Fall 2025

Monte Carlo Helps With **Uncertainty** [WSJ 2016]









Why is there uncertainty?

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion

- Engineering — variability of system parameters, sometimes modeled by Gaussian processes

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion

- Engineering — variability of system parameters, sometimes modeled by Gaussian processes

- Image rendering — can only trace some rays, must rely on a finite sample

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion
- Engineering — variability of system parameters, sometimes modeled by Gaussian processes
- Image rendering — can only trace some rays, must rely on a finite sample
- Bayesian inference — the posterior probability distribution is a combination of a prior and what is learned from data

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion

- Engineering — variability of system parameters, sometimes modeled by Gaussian processes

- Image rendering — can only trace some rays, must rely on a finite sample

- Bayesian inference — the posterior probability distribution is a combination of a prior and what is learned from data

- Neural networks — many parameters need to be tuned, but one cannot search in all possible directions

Why is there uncertainty?

- Finance — market forces, often modeled by stochastic processes driven by Brownian motion
- Engineering — variability of system parameters, sometimes modeled by Gaussian processes
- Image rendering — can only trace some rays, must rely on a finite sample
- Bayesian inference — the posterior probability distribution is a combination of a prior and what is learned from data
- Neural networks — many parameters need to be tuned, but one cannot search in all possible directions
- Queues — arrival times and service times of customers

How is this expressed quantitatively?

$$Y = \text{random variable denoting quantity of interest} = \begin{cases} \text{option payoff} \\ \text{fluid pressure} \\ \text{pixel intensity} \\ \text{statistical model parameter} \\ \text{neural network parameter} \\ \text{service time} \end{cases}$$

$$= f(\boldsymbol{X}), \text{ where}$$

$$\boldsymbol{X} = \text{multivariate random variable with a simpler distribution}$$

We want to estimate the mean, variance, quantile, or probability distribution of Y using sample versions

Are we there yet?

You are visiting your friend and it will require

- A 5 minute walk to the ‘L’ station
- Waiting for the train, which arrives every 20 minutes
- Traveling 35 minutes by ‘L’
- Catching a taxi at the ‘L’ destination
  - There is a 20% chance that the car is waiting for you
  - Otherwise the average wait time is 10 minutes
- A 12 minute taxi ride

How long should you plan for the trip to take?

Let’s look at this Jupyter Notebook AreWeThereYet on the class website

Let me know your degree program?

- Go to menti.com
- Use code 6222 6078

Why should you attend synchronously?

- You will be better keep pace?
- You will get real time answers to questions?
- You can influence the pace and direction of the course?
- You can help your peers learn and benefit from them—partake in a leaning community
- To help me know you, in case you want a reference, or to add me to your LinkedIn network

Review of probability

Let Y be a random variable with a sample space $\mathscr{Y} \subseteq \mathbb{R}$

$\mathbb{P}(Y \in \Omega)$ means the probability of the event Ω

F is the cumulative distribution function $F(y) := \mathbb{P}(Y \le y), \; y \in \mathscr{Y}$

Q is the quantile function $Q(p) := \inf\{y \in \mathscr{Y} : F(y) \ge p\}, \quad 0 < p < 1$

$\mathbb{E}[g(Y)] := \int_{\mathscr{Y}} g(y) \, \mathrm{d}F(y)$ is the expectation of $g(Y)$

discrete	continuous
ϱ is the probability mass function	ϱ is the probability density function
$\varrho(y) := \mathbb{P}(Y = y)$	$\varrho(y) := F'(y)$
$\mathbb{E}[g(Y)] = \sum_{y \in \mathscr{Y}} g(y)\,\varrho(y)$	$\mathbb{E}[g(Y)] = \int_{y \in \mathscr{Y}} g(y)\,\varrho(y)\,\mathrm{d}y$

Moments

$$\mu := \mathbb{E}(Y) \text{ is the mean of } Y$$

$$\sigma^2 = \mathrm{var}(Y) := \mathbb{E}[(Y-\mu)^2] \text{ is the variance of } Y$$

$$\sigma \text{ is the standard deviation}$$

$$\mathrm{cov}(\boldsymbol{Y}) := \big(\mathbb{E}[(Y_j-\mu_j)(Y_k-\mu_k)]\big)_{j,k=1}^{d}$$

is the covariance matrix of the random vector $\boldsymbol{Y}$

Binomial distribution

This is a coin-flipping distribution with the probability p of heads

$$Y \sim \text{Binomial}(n, p) \qquad \texttt{scipy.stats.binom(n, p)}$$

$$\varrho(y) := \mathbb{P}(Y = y) = \binom{n}{y} p^y (1-p)^{n-y}, \quad y = 0, 1, \ldots, n$$

$$F(y) = \mathbb{P}(Y \le y) = \sum_{j=0}^{\lfloor y \rfloor} \binom{n}{j} p^j (1-p)^{n-j}$$

$$Q(p) = \inf\{y \in \{0, 1, \ldots, n\} : F(y) \ge p\}$$

$$\mu = \mathbb{E}(Y) = np \qquad \sigma^2 = \text{var}(Y) = np(1-p)$$

The Bernoulli distribution is the case $n = 1$ and $\mu = p$

Exponential distribution

This distribution models the waiting time for the next customer or taxi, assuming that the time to the next one is independent of how long you have waited so far

$$Y \sim \text{Exponential}(\lambda) \qquad \texttt{scipy.stats.expon(loc=0, scale=}1/\lambda\texttt{)}$$

$$\varrho(y) := \begin{cases} 0, & -\infty < y < 0 \\ \lambda \exp(-\lambda y), & 0 \leq y < \infty \end{cases}$$

$$F(y) = \mathbb{P}(Y \leq y) = \begin{cases} 0, & -\infty < y < 0 \\ 1 - \exp(-\lambda y), & 0 \leq y < \infty \end{cases}$$

$$Q(p) = \frac{-\log(1-p)}{\lambda}, \qquad 0 < p < 1$$

$$\mu = \mathbb{E}(Y) = 1/\lambda \qquad \sigma^2 = \text{var}(Y) = 1/\lambda^2$$

Multivariate normal/Gaussian distribution

Used in finance and uncertainty quantification

$$Y \sim \mathcal{N}(\mu, \Sigma) \quad \text{is a } d\text{-dimensional random vector}$$

`scipy.stats.multivariate_normal(mean, cov)`

$\mu =$ mean

$\Sigma =$ covariance matrix

$$\varrho(\boldsymbol{y}) := \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\tfrac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu})^\top \Sigma^{-1}(\boldsymbol{y}-\boldsymbol{\mu})\right)$$

$$F(\boldsymbol{y}) = \int_{(-\infty,\boldsymbol{y})} \varrho(\boldsymbol{z})\,\mathrm{d}\boldsymbol{z}$$

Sampling

Let Y be be a random variable, often $Y = f(\boldsymbol{X})$

$Y_1, Y_2, \ldots$ be a sample

not necessarily random or independent and identically distributed (IID)

$$\widehat{\mu}_n := \frac{1}{n}\sum_{i=1}^{n} Y_i \text{ is the sample mean}$$

$$\widehat{\sigma}_n^2 := \frac{1}{n-1}\sum_{i=1}^{n} (Y_i - \widehat{\mu}_n)^2 \text{ is the sample variance}$$

Under random sampling

Let Y be be a random variable, often $Y = f(\boldsymbol{X})$

$$Y_1, Y_2, \ldots \sim Y$$

$$\mathbb{E}(\widehat{\mu}_n) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(Y_i) = \mu \text{ so } \widehat{\mu}_n \text{ is unbiased}$$

If in addition, $Y_1, Y_2, \ldots$ are uncorrelated, then

$$\mathbb{E}(\widehat{\sigma}_n^2) = \sigma^2 \text{ so } \widehat{\sigma}_n^2 \text{ is unbiased}$$

$$\operatorname{mse}(\widehat{\mu}_n) := \mathbb{E}[(\mu - \widehat{\mu}_n)^2] = \operatorname{var}(\widehat{\mu}_n) = \frac{\sigma^2}{n}, \qquad \operatorname{rmse}(\widehat{\mu}_n) = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem

Let Y be be a random variable, often $Y = f(\boldsymbol{X})$

$$Y_1, Y_2, \ldots \overset{\text{IID}}{\sim} Y$$

The distribution of

$$\frac{\widehat{\mu}_n - \mu}{\sigma/\sqrt{n}} \text{ approaches } \mathcal{N}(0,1) \text{ as } n \to \infty$$

which means that

$$\widehat{\mu}_n \mathrel{\dot\sim} \mathcal{N}(\mu, \sigma^2/n) \text{ for large } n$$

Monte Carlo estimates of properties of Y

Let Y be a random variable and let $Y_1, Y_2, \ldots$ be an IID random sample

Population Quantity	Sample Approximation
mean μ	$\widehat{\mu}_n \pm 2.58\widehat{\sigma}^2_{n_0}$, `np.mean(data)`
variance σ^2	$\widehat{\sigma}^2_n$, `np.var(data, ddof=1)`
cumulative distribution function F	empirical distribution function $\widehat{F}_n$ `statsmodels.distributions.empirical_distribution.ECDF(data)`
quantile function Q	empirical quantile function $\widehat{Q}_n$ `np.quantile(data, probabilities)`
probability density or mass function ϱ	historgram, `np.histogram(data, bins = "auto")` kernel density estimator, `scipy.stats.gaussian_kde(data)`

For μ we have an interval estimator, but for the others typically point estimators

Are we there yet?

You are visiting your friend and it will require

- A 5 minute walk to the ‘L’ station
- Waiting for the train, which arrives every 20 minutes
- Traveling 35 minutes by ‘L’
- Catching a taxi at the ‘L’ destination
  - There is a 20% chance that the car is waiting for you
  - Otherwise the average wait time is 10 minutes
- A 12 minute taxi ride

How long should you plan for the trip to take?

Let’s look at this Jupyter Notebook AreWeThereYet on the class website

Let me know your professional aspirations?

- Go to menti.com
- Use code 6222 6078

Conditional probability

Let (Y, Z) be be a vector of random variables

$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Y = y \,\&\, Z = z)}{\mathbb{P}(Z = z)}$$

$$\mathbb{P}(Y = y \mid Z = z) = \frac{\mathbb{P}(Z = z \mid Y = y)\mathbb{P}(Y = y)}{\mathbb{P}(Z = z)}$$ Bayes' Theorem

$$\varrho_{Y|Z}(y \mid z) = \frac{\varrho_{Y,Z}(y, z)}{\varrho_Z(z)}$$

$$\mathbb{E}(Y) = \mathbb{E}_Z\left[\mathbb{E}_Y(Y \mid Z)\right]$$

$$\mathrm{var}(Y) = \mathbb{E}_Z[\mathrm{var}_Y(Y \mid Z)] + \mathrm{var}_Z[\mathbb{E}_Y(Y \mid Z)]$$

Conditional Monte Carlo

Let $(Y = f(\boldsymbol{X}_{1:d}), \boldsymbol{X}_{1:d})$ be be a vector of random variables

Let $\boldsymbol{Z} = \boldsymbol{X}_{2:d}$

If $g(\boldsymbol{X}_{2:d}) = \mathbb{E}_Y(Y \mid \boldsymbol{X}_{2:d})$ has an analytic form, then

$\mu = \mathbb{E}(Y) = \mathbb{E}_{\boldsymbol{X}_{2:d}}[g(\boldsymbol{X}_{2:d})]$ and $\operatorname{var}(Y) \geq \operatorname{var}_{\boldsymbol{X}_{2:d}}[g(\boldsymbol{X}_{2:d})]$

If $h(y, \boldsymbol{X}_{2:d}) = \varrho_{Y|\boldsymbol{X}_{2:d}}(y \mid \boldsymbol{x}_{2:d})$ has an analytic form, then

$\varrho_Y(y) = \mathbb{E}_{\boldsymbol{X}_{2:d}}[h(y, \boldsymbol{X}_{2:d})]$ probability density expressed as expectation

Big Ideas

- Sample quantities, often using IID samples, used to estimate population properties of a random variable with a complex distribution, including
  - Means, variances, covariances
  - Probability density/mass functions, cumulative distribution functions, and quantile functions
- For means we have interval estimators, which indicate the uncertainty in our estimates, using the CLT
- For the other properties we have point estimators, i.e., only one approximate value for each thing to be estimated